Inner Product Spaces I–II

From the class book: Chapter 7, Sections 7.1–7.4

1. Key definitions

Inner product. On a vector space V over $F = \mathbb{R}$ or $\mathbb{C}$, a map $\langle \cdot, \cdot \rangle$ that is linear in the first argument, conjugate symmetric ($\langle v, u \rangle = \overline{\langle u, v \rangle}$) and positive ($\langle v, v \rangle > 0$ for $v \neq 0$). The norm is $\|v\| = \sqrt{\langle v, v \rangle}$.

Orthogonality. $u \perp v$ if $\langle u, v \rangle = 0$. For $S \subseteq V$, $S^{\perp} = \{v : \langle v, s \rangle = 0 \ \forall s \in S\}$ is always a subspace. An orthonormal set is a set of pairwise orthogonal unit vectors.

- Standard examples: $\sum x_i y_i$ on $\mathbb{R}^n$, $\sum x_i \bar{y}_i$ on $\mathbb{C}^n$, $\int_a^b fg$ on $C[a, b]$, $\operatorname{tr}(B^t A)$ on $M_n(\mathbb{R})$.
- $\langle x, y \rangle = y^t A x$ on $\mathbb{R}^n$ is an inner product iff A is symmetric positive definite.

2. Main results

Theorem (Norm inequalities and identities)

- Cauchy–Schwarz: $|\langle u, v \rangle| \leq \|u\| \, \|v\|$, with equality iff u, v are dependent.
- Triangle inequality: $\|u + v\| \leq \|u\| + \|v\|$.
- Pythagoras: $u \perp v \Rightarrow \|u + v\|^2 = \|u\|^2 + \|v\|^2$ (the converse holds over $\mathbb{R}$).
- Parallelogram law: $\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2$. A norm comes from an inner product iff it satisfies this law; $\| \cdot \|_1$, $\| \cdot \|_\infty$ do not.
- Polarization (real case): $\langle u, v \rangle = \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2)$.

Theorem (Orthonormal bases and projections)

- Non-zero orthogonal vectors are independent. If $\{e_i\}$ is an orthonormal basis, $v = \sum \langle v, e_i \rangle e_i$ and $\|v\|^2 = \sum |\langle v, e_i \rangle|^2$.
- For a finite-dimensional subspace W: $V = W \oplus W^{\perp}$, $(W^{\perp})^{\perp} = W$, $\dim W^{\perp} = \dim V - \dim W$.
- If $\{e_1, \ldots, e_k\}$ is an orthonormal basis of W, the projection is $P_W v = \sum \langle v, e_i \rangle e_i$, and $P_W v$ is the point of W nearest to v: $d(v, W) = \|v - P_W v\|$.
- Distance to a hyperplane $a \cdot x = 0$ in $\mathbb{R}^n$: $|a \cdot v| / \|a\|$.

Matrix	Definition	Eigenvalues
Hermitian / real symmetric	$A^* = A$	real
skew-Hermitian	$A^* = -A$	in $i\mathbb{R}$
unitary / orthogonal	$A^*A = I$	$\lvert\lambda\rvert = 1$
positive definite	$A^* = A$, $\langle Av, v\rangle > 0$	> 0
orthogonal projection	$P^2 = P = P^*$	$0, 1$

3. Methods

Method: Gram–Schmidt.

- (1) $v_1 = w_1$, and $v_j = w_j - \sum_{i<j} \frac{\langle w_j, v_i\rangle}{\langle v_i, v_i\rangle} v_i$.
- (2) Normalize: $e_j = v_j / \|v_j\|$. If some $v_j = 0$, drop w_j (it depends on the earlier vectors).
- (3) On $[-1, 1]$, Gram–Schmidt on $1, x, x^2$ gives $1,\ x,\ x^2 - \frac{1}{3}$ (Legendre polynomials up to scaling).

Method: is a formula an inner product?

- For a bilinear form on $\mathbb{R}^n$, write its symmetric matrix A and test positive definiteness (all leading principal minors > 0).
- For 2×2: $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$ is positive definite iff $a > 0$ and $ac - b^2 > 0$.

4. Exam shortcuts and common traps

- In $M_n(\mathbb{R})$ with $\operatorname{tr}(B^t A)$: symmetric $\perp$ skew-symmetric, and $M_n = \mathbf{Sym} \oplus \mathbf{Skew}$ orthogonally.
- $\|u\| = 3$, $\|v\| = 4$, $\|u + v\| = 5 \Rightarrow u \perp v$ and $\|u - v\| = 5$.
- Hermitian + unitary $\Rightarrow A^2 = I$; unitary and triangular $\Rightarrow$ diagonal.
- **Traps:** $\operatorname{tr}(AB)$ is not an inner product on $M_n(\mathbb{R})$ (negative on skew matrices); $x_1y_1 - x_2y_2$ fails positivity; over $\mathbb{C}$, $\|u + v\|^2 = \|u\|^2 + \|v\|^2$ gives only $\operatorname{Re}\langle u, v\rangle = 0$.

5. Solved examples

Example 1. For which α is $\langle (x_1, y_1), (x_2, y_2)\rangle = x_1x_2 - \alpha(x_1y_2 + x_2y_1) + 4y_1y_2$ an inner product on $\mathbb{R}^2$?

Solution

- Its matrix is $\begin{pmatrix} 1 & -\alpha \\ -\alpha & 4 \end{pmatrix}$.
- Positive definite iff $4 - \alpha^2 > 0$, i.e. $\lvert\alpha\rvert < 2$.

Example 2. Apply Gram–Schmidt to $1, x, x^2$ with $\langle p, q\rangle = \int_{-1}^{1} pq$.

Solution

- $v_1 = 1$; $\langle x, 1\rangle = 0$, so $v_2 = x$.
- $\langle x^2, 1\rangle = \frac{2}{3}$, $\langle 1, 1\rangle = 2$, $\langle x^2, x\rangle = 0$: $v_3 = x^2 - \frac{1}{3}$.

Example 3. Find the squared distance from $(1, 2, 3)$ to the plane $x + y + z = 0$.

Solution

- The plane is $a^{\perp}$ with $a = (1, 1, 1)$.
- $d^2 = \frac{(a\cdot v)^2}{\|a\|^2} = \frac{36}{3} = 12$.

Example 4. Project $v = (1, 0, 0, 0)$ onto $W = \operatorname{span}\{(1, 1, 0, 0), (0, 0, 1, 1)\}$ in $\mathbb{R}^4$.

Solution

- Orthonormal basis: $e_1 = \frac{1}{\sqrt{2}}(1, 1, 0, 0)$, $e_2 = \frac{1}{\sqrt{2}}(0, 0, 1, 1)$.
- $\langle v, e_1\rangle = \frac{1}{\sqrt{2}}$, $\langle v, e_2\rangle = 0$, so $P_W v = (\frac{1}{2}, \frac{1}{2}, 0, 0)$ and $d(v, W)^2 = \frac{1}{2}$.

Example 5. In $M_2(\mathbb{R})$ with $\langle A, B\rangle = \operatorname{tr}(B^t A)$, find the orthogonal complement of the symmetric matrices.

Solution

- If S is symmetric and K skew, $\operatorname{tr}(K^t S) = -\operatorname{tr}(KS) = 0$, so skew $\subseteq \mathrm{Sym}^{\perp}$.
- Dimensions: $4 - 3 = 1 = \dim \mathrm{Skew}$. So $\mathrm{Sym}^{\perp} = \mathrm{Skew}$.

Example 6. In a real inner product space, $\|u\| = 3$, $\|v\| = 4$, $\|u + v\| = 5$. Find $\langle u, v\rangle$ and $\|u - v\|$.

Solution

- $25 = 9 + 16 + 2\langle u, v\rangle \Rightarrow \langle u, v\rangle = 0$.
- Parallelogram law: $\|u - v\|^2 = 2\cdot 9 + 2\cdot 16 - 25 = 25$, so $\|u - v\| = 5$.